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Pedestrian Crossing Prediction for Automated Vehicles: A Geometry-aware Method Using Surface Normal

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Abstract

Accurate prediction of pedestrian crossing intention is critical for the safety of automated vehicles. Existing approaches mainly rely on visual appearance and scene context derived from raw images and semantic segmentation. We propose a geometry-aware framework that incorporates surface normal maps encoding pixel-level 3D surface orientation. To integrate heterogeneous information, we extend a multi-stream GRU-attention architecture with surface-normal features. The visual branch combines surface normals, local context, and segmentation overlays, while the non-visual branch includes body pose, bounding boxes, and ego-vehicle speed. Extensive ablation studies on the PIE dataset demonstrate that surface normal maps provide consistent performance gains. When combined with contextual features, they improve F1 score and AUC by 3.85% and 2.35%, respectively, and yield more stable predictions up to two seconds before crossing events. These results highlight the importance of geometry-informed features as an effective complement to conventional visual inputs.

CCS Concepts

• Computing methodologies → Scene understanding.

Keywords

Pedestrian Crossing Intention Prediction, Automated Vehicles, Surface Normal, Feature Fusion, Context-aware prediction

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1 Introduction

Pedestrian crossing prediction, also known as pedestrian intention prediction (PIP) models, aim at forecasting whether a pedestrian will cross the road in the near future. This automated driving function is a key element in the enhancement of intelligent and safe autonomous driving systems [27]. In complex urban environments, vehicles often encounter pedestrians whose actions must be accurately anticipated to ensure safe and timely manoeuvres. As one of the most vulnerable and unpredictable road users, pedestrians play a significant role in traffic-related incidents. Reliable prediction systems must therefore analyse and integrate pedestrian motion cues, environmental context, and interactions with the ego-vehicle to effectively infer crossing intentions.

Previous studies have utilised visual and non-visual cues such as pedestrian bounding boxes, body pose keypoints, and vehicle dynamics to infer pedestrian crossing behaviour [1, 11, 26]. However, many existing approaches rely heavily on raw frame features or semantic segmentation inputs, both prone to degradation under challenging conditions such as poor lighting or visual ambiguity [24, 25]. Current approaches often lack an explicit representation of physical scene elements, such as curbs, pavements, and road surfaces, which are essential factors influencing pedestrian decision-making. While appearance and semantic cues capture visual context, they provide limited insight into the physical layout of the scene, which strongly influences pedestrian crossing behaviour (e.g., the spatial separation between road and sidewalk). A geometry-aware representation helps the model understand which parts of the scene are walkable or suitable for crossing.

To address these limitations, we investigate the integration of surface normal maps as a complementary geometry-aware modality for pedestrian crossing prediction. Rather than proposing a new prediction architecture, our objective is to evaluate whether explicit geometric scene representations provide complementary information beyond conventional appearance- and semantics-based features. Surface normal maps represent the per-pixel 3D orientation of visible scene surfaces and thus encode scene structure rather than appearance. Although they are estimated from RGB frames using DSINE [2], they capture geometric cues that are relatively less sensitive to lighting, shadows, or texture variations [14]. Surface normals have been successfully employed in various tasks, including 3D reconstruction and affordance prediction. However, their potential for pedestrian crossing prediction remains unexplored.

Unlike depth maps or Lidar-derived geometry, surface normals offer a compact, camera-derived representation of surface orientation that can be estimated directly from monocular videos.

Inspired by prior multi-stream designs [1, 11, 26], the proposed pedestrian crossing prediction model processes six input modalities: local context, semantic segmentation (via overlaid raw-frame-segmentation maps), surface normal maps, bounding box, body pose, and ego-vehicle speed. Each modality is temporally modelled using a dedicated GRU-attention stream to capture its temporal dynamics, and the resulting features are fused using a hybrid late-stage strategy. Experiments on the PIE dataset [19], a widely used benchmark dataset for pedestrian behaviour analysis, demonstrate that incorporating surface normal maps consistently improves performance across multiple metrics. Notably, surface normals enhance the consistency of model predictions over different time-to-event (TTE) horizons. We adopt Yang et al. [26] multi-stream design not to propose a new architecture, but to systematically evaluate the influence of adding a geometry-informed modality under a controlled, established baseline. Our key contributions are as follows:

- We introduce surface normal maps as a complementary geometry-aware representation for pedestrian crossing prediction, representing the first application of this modality in this context.
- We demonstrate that integrating surface normal maps into an established multi-stream GRU-attention framework consistently improves pedestrian crossing prediction performance across Accuracy, F1 score, and AUC metrics.
- We conduct comprehensive ablation studies and temporal analyses showing that incorporating surface normals improves model robustness across longer time-to-event intervals.

By incorporating surface normal information into crossing prediction, we introduce a complementary geometric cue that enhances spatial understanding of urban scenes and supports more reliable decision-making for autonomous systems.

2 Related Work

Early approaches to pedestrian crossing prediction primarily relied on handcrafted features or convolutional neural networks (CNNs) applied to RGB data [20, 21]. While effective in capturing visual appearance, these methods often overlook non-visual signals such as pedestrian dynamics or vehicle motion. Subsequent works addressed this limitation by incorporating additional modalities. For instance, Kotseruba et al. [13] introduced the PCPA framework, which fused image features with contextual and dynamic cues. Cadena et al. proposed the Pedestrian Graph [4] and its enhanced version [3], integrating pose keypoints, image context, and ego-vehicle speed for richer scene understanding. Similarly, Liu et al. [15] modelled interactions between pedestrians and other road users using a graph convolutional network (GCN), capturing both spatial relationships and temporal dependencies in dynamic scenes.

With the advent of transformer-based architectures, several models have advanced the field by capturing long-range dependencies through self-attention mechanisms. TrEP [28] models temporal relations in pedestrian behaviour while accounting for uncertainty

via evidential learning, whereas TAMformer [17] employs a multi-modal transformer to forecast future pedestrian actions across multiple anticipation horizons. PedFormer [18] introduces cross-modal attention and gated multi-task learning for joint trajectory and intention prediction.

Alongside these developments, multi-modal fusion techniques have also been explored. Ham et al. [11] proposed MCIP, a multi-stream network that processes heterogeneous features in parallel to exploit their complementary properties. PIP-Net [1] integrates spatial and kinematic inputs using recurrent units and temporal attention, and introduces local motion flow and categorical depth features for improved visual encoding.

Despite these advancements, most state-of-the-art models rely primarily on RGB appearance or 2D semantic segmentation to represent scene context. However, such representations are sensitive to adverse illumination, reflections, or low-texture regions [14], and often lack explicit 3D geometric understanding.

Surface normal maps provide an explicit representation of scene geometry by encoding the per-pixel orientation of visible surfaces. These geometric cues capture structural properties of the environment, such as surface layout, boundaries, and spatial configuration, that are directly relevant to pedestrian movement and crossing behaviour. While surface normal estimation has been widely studied in geometric scene understanding and reconstruction tasks [9, 10, 12, 22], its potential role in behavioural reasoning for pedestrian crossing prediction remains largely unexplored.

Existing pedestrian crossing prediction approaches primarily rely on appearance-based or motion-related features, which may not explicitly represent the physical structure of the environment, such as road boundaries, sidewalks, or curb geometry. This work addresses this gap by introducing surface normal maps as a complementary geometric modality for pedestrian crossing prediction. The proposed approach is motivated by the hypothesis that incorporating explicit scene geometry can support more reliable reasoning about pedestrian behaviour, particularly in complex urban environments where spatial layout and surface configuration influence crossing decisions.

3 Methodology

We address the task of pedestrian crossing prediction as follows: given a sequence of $m+1$ consecutive frames captured by a forward-facing, vehicle-mounted camera, the objective is to predict whether a target pedestrian i will begin crossing the road at time t , which is also known as crossing event time. A critical aspect of this prediction is the Time-to-Event (TTE), defined as the interval between the last frame of the model's observation window and the actual start of the crossing event. Our model aims to make this prediction for TTE values typically ranging from 30 to 60 frames (1 to 2 seconds) [20]. We adopt an established inspired by prior multi-stream designs [1, 11, 26] enabling a controlled evaluation of the contribution of surface normal maps while maintaining a consistent architectural baseline. The proposed model, shown in Figure 1, utilises three primary visual input streams (local context, semantic segmentation, and surface normal maps) and three non-visual modalities (body pose, bounding box, and ego-vehicle speed). The main model also uses an overlaid representation for the semantic

segmentation stream. Below, we provide more details about the Visual and non-visual Inputs.

3.1 Visual Inputs

The visual branch processes three distinct types of visual information to capture complementary cues:

Local Context (L^i) is obtained from the RGB frame image, which is cropped around each pedestrian's bounding box, expanded by a factor of 1.5, and subsequently resized to 224×224 pixels. Local context represents the local appearance and surrounding scene elements of the target pedestrian. The temporal sequence of local context frames is denoted as $L^i = \{l_{t-m}^i, \dots, l_t^i\}$.

Semantic Segmentation information is the second visual input. Raw semantic segmentation masks (S^i) are generated using DeepLabV3 [6], pretrained on the Cityscapes dataset [8]. These masks represent scene semantics, including roads, pavements, and vehicles, and are computed from the entire frame. The temporal sequence is defined as $S^i = \{s_{t-m}^i, \dots, s_t^i\}$.

In our main proposed model, to better balance appearance and semantic cues while preserving spatial coherence, we utilise **Overlaid RGB-Segmentations (SO^i)** as the input for this stream. SO^i is a composite input created by overlaying the raw semantic mask onto the full frame with 50% opacity. The resulting sequence is denoted as $SO^i = \{so_{t-m}^i, \dots, so_t^i\}$.

Blending semantic masks with raw frames enhances spatial understanding by retaining low-level texture and high-level structural information. The input SO^i (or S^i in ablation studies; see Section 4) is resized to 224×224 before being fed to its dedicated stream.

Surface Normal Maps (N^i) are estimated using the approach of Bae et al. [2], encoding per-pixel 3D surface orientation (e.g., road planarity and curb edges). Surface normals provide geometry-aware priors that are comparatively less sensitive to lighting variations and are computed from the entire frame. The temporal sequence of surface normals is defined as $N^i = \{n_{t-m}^i, \dots, n_t^i\}$.

Surface normals maintain spatial consistency through piecewise smoothness and structural regularity, enabling the model to distinguish planar surfaces (e.g., roads) from vertical ones (e.g., curbs or pedestrians). Sharp changes in normals at geometric boundaries further refine segmentation masks, ensuring alignment with the physical environment. While normals lack semantic identity, their fusion with semantic features creates a physically coherent representation that enhances generalisation in unpredictable urban settings.

Each stream is processed through a CNN-GRU-attention pipeline (Section 3.3) to generate modality-specific embeddings.

3.2 Non-visual Inputs

In parallel with the visual branch, the non-visual branch processes features related to pedestrian motion and ego-vehicle dynamics. These include the target pedestrian's bounding box, body pose keypoints, and the speed of the ego-vehicle.

Bounding Box (B^i): Each frame in the PIE dataset includes a 2D bounding box defined as $b_t^i = [x_1, y_1, x_2, y_2]$ for the target pedestrian. The temporal sequence of these bounding boxes, $B^i = \{b_{t-m}^i, \dots, b_t^i\}$, is used as an input to the model. For each b_t^i , the coordinates are normalised relative to the frame resolution before

being fed to the model. This sequence encodes the spatial position and trajectory of the target pedestrian over the observed window.

Body Pose (P^i): We use OpenPose [5] to extract 18 body keypoints per frame, resulting in a 36-dimensional pose vector. These keypoints are obtained by applying the pose estimation model to the local context L^i . The temporal pose sequence $P^i = \{p_{t-m}^i, \dots, p_t^i\}$ captures the posture dynamics of pedestrians over time.

Ego-Vehicle Speed (V): The sequence $V = \{v_{t-m}, \dots, v_t\}$ contains frame-wise speed values for the ego-vehicle, where each v_t represents the vehicle speed at frame t . This modality provides motion context for pedestrian-vehicle interaction.

Each modality is encoded through a dedicated GRU-attention stream, and the resulting representations are fused using a late-fusion strategy for binary classification.

3.3 Model Architecture and Feature Fusion

Our multi-stream architecture, inspired by [1, 11, 26], processes visual and non-visual modalities via dedicated streams, each employing a GRU-attention mechanism for temporal modelling, followed by a hybrid fusion strategy.

Feature Encoders. Visual streams (L^i, SO^i, N^i) are processed frame-wise by a VGG19 [23] backbone. Input frames ($224 \times 224 \times 3$) yield 512-dim feature vectors per frame from the fourth max-pooling layer (via global average pooling), forming sequences of length $m+1$. These sequences, and also non-visual sequences (P^i, B^i, V), are then fed into modality-specific Gated Recurrent Units (GRUs) [7] (hidden size 256) to capture temporal dynamics, producing hidden state sequences $H = \{h_1, \dots, h_{m+1}\}$.

Temporal Attention. A temporal attention mechanism [16] weights the GRU hidden states H for each stream. Given H and its final state $h_e = h_{m+1}$, attention weights α_s are computed as $\alpha_s = \text{softmax}(h_e^T W_s \tilde{h}_s)$ over all $\tilde{h}_s \in H$. The context vector $h_c = \sum \alpha_s \tilde{h}_s$ and h_e are combined: $\text{Atten} = \tanh(W_c [h_c; h_e])$, yielding a fixed-size (e.g., 256-dim) embedding per stream. W_s, W_c are learnable parameters.

Feature Fusion and Prediction. We follow the hybrid fusion strategy introduced by Yang et al. [26], which separates inputs into two branches: a non-visual branch and a visual branch.

In the non-visual branch, we hierarchically fuse pose keypoints, bounding boxes, and ego-vehicle speed. First, pose sequences are passed through a GRU encoder. The resulting embeddings are concatenated with the bounding box sequence and fed into a second GRU. This output is further concatenated with the ego-vehicle speed sequence and processed by a final GRU. A temporal attention module (as described above) is applied to the last GRU's output to produce a 256-dimensional non-visual embedding V_{kin} .

In the visual branch, each stream, local context, semantic segmentation, and surface normal maps, is encoded independently using the CNN-GRU-Temporal Attention pipeline described previously, each yielding a 256-dimensional embedding. These three embeddings are then concatenated to form the visual embedding of $V_{\text{vis}} \in \mathbb{R}^{768}$.

Finally, the non-visual embedding $V_{\text{kin}} \in \mathbb{R}^{256}$ and the visual embedding $V_{\text{vis}} \in \mathbb{R}^{768}$ are fused using a modality attention mechanism before the final prediction layer. This mechanism learns to scale the

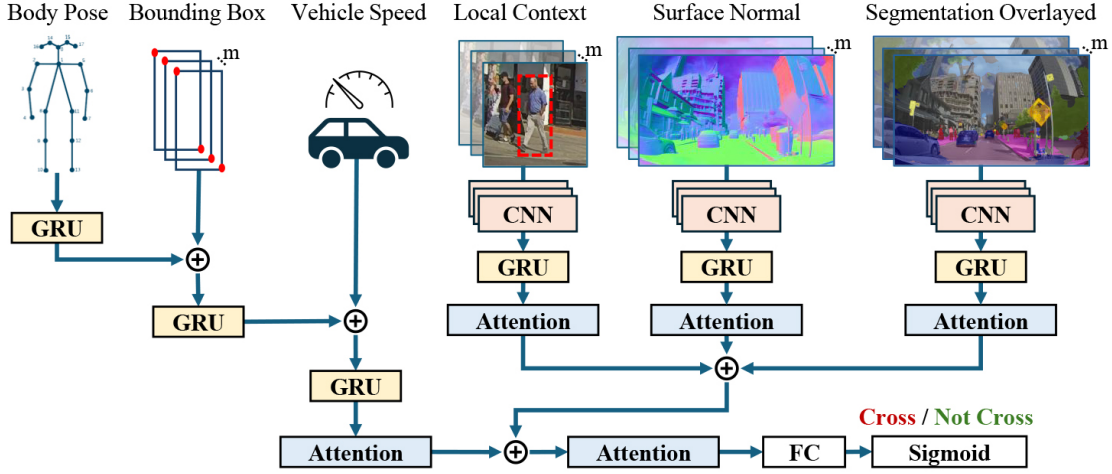


Figure 1: Overview of the proposed multi-stream architecture for pedestrian crossing prediction, including visual and kinematic branches, attention-based feature summarisation, and late fusion.

contribution of the aggregated visual and non-visual information. The fused vector is then passed through a fully connected (FC) layer with sigmoid activation to predict the crossing label.

4 Experiments and Results

We conduct a series of experiments to evaluate the impact of incorporating surface normal maps into an existing pedestrian crossing prediction model. Our evaluation focuses on three aspects: (1) comparing various visual feature configurations through ablation studies, (2) analysing the temporal robustness of predictions across time-to-event (TTE) intervals, and (3) evaluating the contribution of surface normals when used alongside appearance- and semantics-based features.

4.1 Experimental Setup

We evaluate our approach on the well known PIE dataset [19] in the research community, which contains over six hours of annotated urban driving footage captured from a vehicle-mounted camera. The dataset includes 1,842 pedestrian instances across diverse scenarios, with 512 labelled as crossing and 1,322 as non-crossing. Each instance is annotated with bounding boxes, crossing behaviour, and ego-vehicle speed.

Following the standard protocol of Yang et al. [26], we use an observation window of $m+1 = 16$ frames at 30 FPS to predict whether the target pedestrian will begin crossing shortly after the final frame. Performance is evaluated using *Accuracy*, *Precision*, *Recall*, *F1 score*, and *Area Under the ROC Curve (AUC)*.

All models are implemented in TensorFlow and trained using the Adam optimiser with a learning rate of 5×10^{-5} and a batch size of 2 for 60 epochs. Binary cross-entropy serves as the loss function. We apply a dropout rate of 0.5 in the attention modules and L2 regularisation ($\lambda = 0.001$) in the final classification layer. All experiments follow the dataset split defined in the PIE benchmark [13].

Table 1: Performance of different visual feature combinations on the PIE dataset. All configurations include pose, bounding box, and vehicle speed.

#	Visual Components	Acc	AUC	F1	Precision	Recall
1	$L + S$ [26]	0.88	0.85	0.78	0.78	0.79
2	$L + SO$	0.89	0.86	0.80	0.79	0.80
3	$L + N$	0.88	0.86	0.80	0.79	0.80
4	$L + S + N$	0.88	0.86	0.79	0.76	0.83
5	$L + SO + N$	0.90	0.87	0.81	0.81	0.82

4.2 Visual Feature Ablation Study

We conduct an ablation study with five model configurations to assess contributions of various visual scene representations, particularly semantic and geometric cues. Our baseline includes essential features widely used in pedestrian prediction: local context (L), body pose (P), bounding boxes (B), and ego-vehicle speed (V). These are kept constant to isolate the impact of varied visual streams. The study focuses on ablating two key additional visual inputs from the broader scene: semantic segmentation and surface normal maps. To isolate the contribution of geometric and semantic visual features, all non-visual modalities (pose, bounding box, and vehicle speed) were kept constant across experiments.

The ablated features are: N (surface normal maps), S (raw segmentation masks), and SO (segmentation masks overlaid on RGB frames with 0.5 opacity).

The ablation results in Table 1 demonstrate the progressive benefit of integrating richer visual cues. *Config #2* ($L + SO$) shows that overlaying semantic masks on RGB frames improves performance (e.g., +0.02 F1 over *Config #1* using raw segmentation masks, likely by providing the VGG19 with both low-level texture and high-level semantic structure, enhancing spatial understanding).

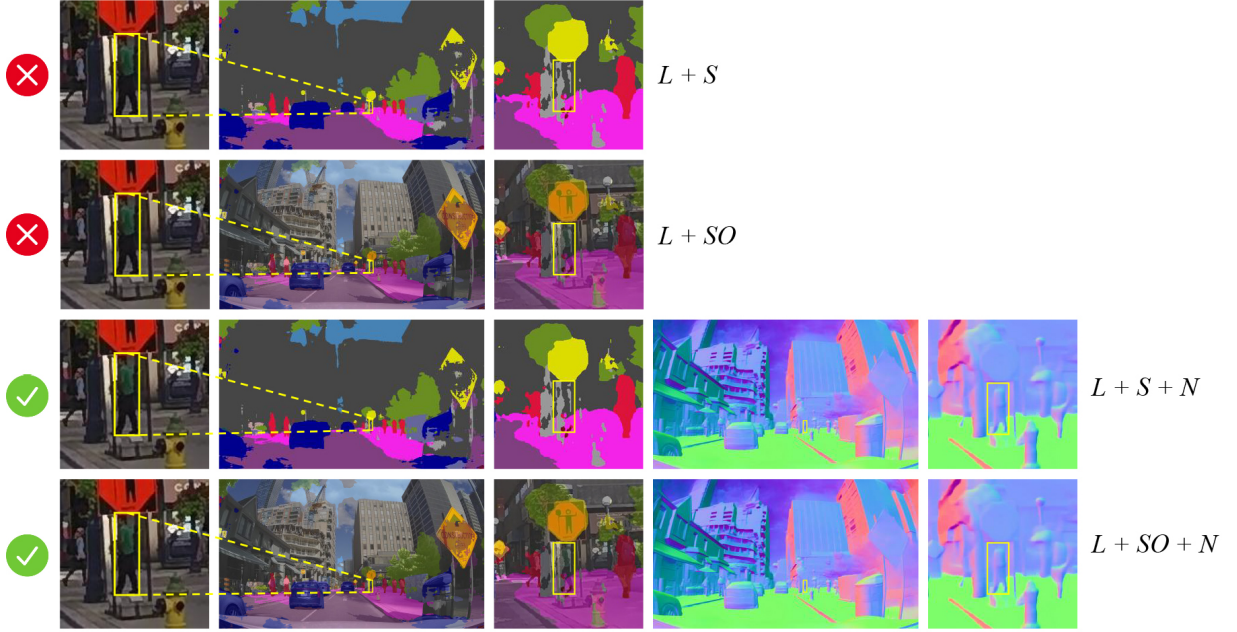


Figure 2: Qualitative results demonstrating how surface normal maps can mitigate segmentation failures in pedestrian crossing prediction.

The introduction of surface normal maps (N) yields further improvements. When used as the sole additional visual stream alongside local context (*Config #3: $L + N$*), surface normals achieve performance comparable or superior to using raw segmentation (*Config #1*) and similar to using overlaid segmentation (*Config #2*), highlighting their substantial standalone informative value (e.g., F1 of 0.80 for *Config #3* vs. 0.78 for *Config #1*).

The most notable improvements occur when surface normals are integrated with the overlaid segmentation maps (*Config #5: $L + SO + N$*). This setup delivers the best overall performance (Accuracy 0.90, AUC 0.87, F1 0.81), surpassing *Config #2* (without normals) in all metrics. This indicates a synergistic relationship, where the strong geometric understanding from surface normals (e.g., explicit curb and road planarity) enhances the rich appearance and semantic information from SO , resulting in a more comprehensive and dependable scene representation.

Interestingly, while adding surface normals to raw segmentation maps (*Config #4: $L + S + N$*) substantially boosts Recall to 0.83 (compared to 0.79 for *Config #1*), it results in a slight decrease in Precision (0.76 vs. 0.78). This indicates that while geometric cues help identify more true crossing instances, potentially resolving ambiguities in poorly segmented regions, they might also contribute to an increase in false positives when combined only with raw segmentation. The superior performance of *Config #5* suggests that the richer overlaid RGB-segmentation (SO) input helps contextualise the geometric cues from surface normal maps more effectively.

Figure 2 presents qualitative examples illustrating how surface normal maps are complementary in scenarios where semantic segmentation fails due to visual ambiguity. In Row 1 ($L + S$), the model fails to predict the pedestrian as crossing. Despite the RGB image

containing rich visual detail, the segmentation map misidentifies the pedestrian, likely due to shadowing and colour similarity with the background, resulting in an incorrect prediction. Row 2 ($L + SO$) overlays the segmentation mask onto the RGB image to improve visibility. While this aids in perceptual clarity, it does not rectify the underlying segmentation error, and the model still misclassifies the crossing.

In contrast, Rows 3 and 4 ($L + S + N$ and $L + SO + N$) utilise surface normal maps, offering illumination-invariant geometric cues that effectively capture the shape and layout of objects. While segmentation struggles to clearly define the pedestrian in these instances, the surface normals maintain the pedestrian’s form, allowing the model to predict crossing behaviour accurately. These examples support our hypothesis that surface normals enhance scene understanding by offering geometry-aware features that compensate for the limitations of appearance- and semantics-based inputs.

4.3 Temporal Performance Analysis

To further analyse model robustness, we evaluate its predictive capability across the different TTEs. Specifically, we assess how performance varies when the 16-frame observation window ends at different proximities (30 to 60 frames, or 1 to 2 seconds) to the actual crossing event. This tests the model’s ability to make reliable predictions based on observations made further in advance of the event. Figure 3 presents Accuracy, F1 score, and AUC for key model configurations across these TTE values.

As depicted in Figure 3, the configuration combining surface normals with the segmentation overlay (*Config #5: $L + SO + N$* , green line) consistently yields the highest or near-highest scores across all metrics and most TTE horizons. Notably, it maintains an

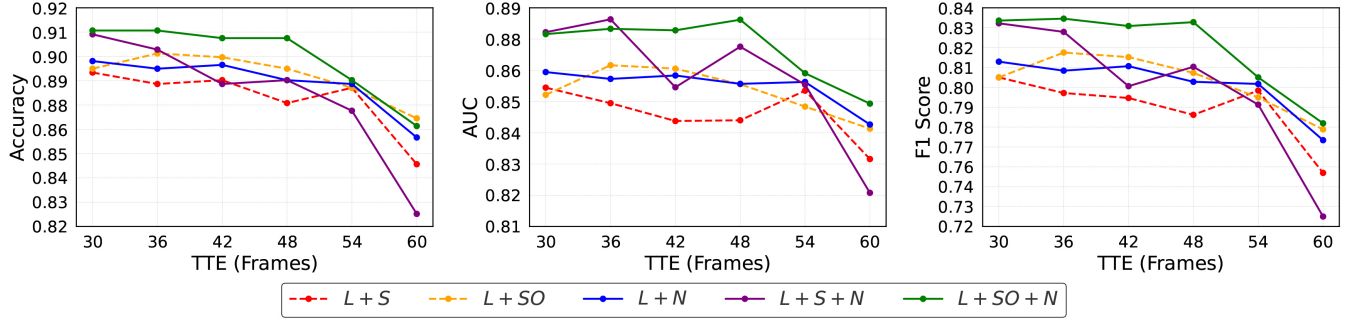


Figure 3: Comparison of model performance over different TTE intervals.

accuracy exceeding 0.90 and an F1 score surpassing 0.82 up to a TTE of 48 frames (1.6 s). Crucially, its AUC exhibits the most graceful degradation, remaining superior to all other configurations across the entire TTE range, particularly at longer TTEs (e.g., maintaining an AUC above 0.85 at TTE = 60 frames). This superior AUC performance indicates a more robust ranking of crossing likelihoods, even when predictions are made earlier before the crossing event. We hypothesise that the rich contextual information derived from the segmentation overlay, combined with the stable geometric cues provided by surface normals, contributes to improved robustness across longer temporal horizons.

In contrast, the model using raw segmentation combined with surface normals (*Config #4: L + S + N*, purple line) demonstrates more rapid performance degradation at longer TTEs. Although competitive in shorter TTEs, its AUC, for instance, drops more dramatically than *Config #5* after TTE = 42 frames, suggesting that inconsistencies in raw semantic masks become more detrimental when observations are further removed from the event.

The model utilising only the segmentation overlay with local context (*Config #2: L + SO*, orange line) also performs competitively, generally outperforming configurations relying solely on raw segmentation (*Config #1*, red dashed line) or solely on surface normals with local context (*Config #3*, blue line) across many TTEs. This validates overlaid RGB-Segmentation as a strong stand-alone visual representation. However, the consistent gap between *Config #2* (orange line) and *Config #5* (green line) underscores the added benefit of surface normals for temporal robustness. Overall, these temporal results confirm that the synergistic combination of appearance (local context), rich semantics (segmentation overlay), and explicit geometric structure (surface normals), as in *Config #5*, leads to more generalisable and reliable pedestrian crossing predictions, particularly in challenging scenarios requiring decisions based on earlier observations (i.e., larger TTEs).

5 Conclusion and Future Work

This work explored the integration of geometry-aware features, specifically surface normal maps, into a pedestrian crossing prediction framework. Extending an existing multi-stream GRU-attention architecture, we incorporated surface normals as a complementary modality alongside RGB appearance, semantic segmentation, pose keypoints, bounding boxes, and ego-vehicle speed.

Comprehensive evaluations on the PIE dataset, including ablation studies and TTE analysis, showed that surface normals consistently improve robustness, particularly in visually ambiguous or structurally complex scenes. The combination of surface normals and overlaid semantic segmentation achieved the most consistent gains across all evaluation metrics, especially at longer prediction horizons, which is critical for AV’s safe and timely manoeuvring. This highlights the importance of encoding geometric structure in a noisy road environment in order to complement traditional appearance and motion cues in crossing prediction tasks. Unlike appearance-based features, surface normals provide fine-grained spatial information that remains stable across lighting and texture changes, offering more reliable inference in challenging urban scenarios.

This study intentionally adopted a pre-existing architecture, including a VGG19 backbone and GRU-based temporal encoders, following Yang et al. [26], but isolated the main part of the model in order to conduct a fair evaluation and comparison of the surface normal maps’ efficiency.

The current pipeline relies on multiple task-specific models (OpenPose, DeepLabV3, and DSINE) for feature extraction, which increases computational cost and limits real-time applicability. Since the primary objective of this work was to evaluate the predictive contribution of geometry-aware representations under a controlled experimental setting, runtime performance was not explicitly evaluated. Future work will investigate lightweight end-to-end architectures and more efficient feature extraction pipelines suitable for real-time autonomous driving applications.

This study focused on evaluating the contribution of surface normal representations estimated from RGB video data. Although surface normals can also be derived from other sensing modalities, such as LiDAR or radar, the present work deliberately adopts a single-modality (video-only) setup to isolate and examine the geometric contribution of surface orientation cues within a vision-centric framework.

In addition, the evaluation was conducted exclusively on the PIE dataset. While PIE is a widely used benchmark for pedestrian behaviour analysis, validating the proposed approach across datasets with different environments, viewpoints, and traffic conditions would provide further insight into its generalisability. Future work can therefore explore cross-dataset validation and investigate the

integration of geometry-aware representations within broader sensing configurations.

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